

DATA

MISSING REPLACEMENT

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MISSING REPLACEMENT

The following concepts will be introduced:

- ✓ **MISSING DATA**

- ✓ **MISSING REPLACEMENT**

- RECORD REMOVAL
- MANUAL IMPUTATION
- GLOBAL CONSTANT
- MEAN/MODE
- CONDITIONAL MEAN
- MOST PROBABLE

MISSING REPLACEMENT

1



The **CUSTOMERS CALLS DATABASE**, provided by the CEO of the **TELECOM COMPANY**, contains customers' call data consisting of many attributes including **CHURN**, a binary attribute representing whether a given **CUSTOMER CHURNED FOR A COMPETITOR OR NOT**.

Account Length	VMail Message	Day Mins	Churn	Intl Calls	Intl Charge	State	Area Code	Phone
128	25	265.1	?	3	2.7	KS	415	382-4657
107	26	161.6	n	3	3.7	OH	415	371-7191
137	0	243.4	n	5	3.29	NJ	415	358-1921
84	0	299.4	n	7	1.78	?	408	375-9999
75	0	166.7	n	3	2.73	OK	415	330-6626
118	0	223.4	y	?	1.7	KS	510	391-8027
121	24	218.2	n	7	2.03	MA		355-9993
147	0	157	y	6	1.92	MO	415	329-9001
117	0	184.5	n	4	2.35	KS	408	335-4719
141	37	258.6	n	5	3.02		415	330-8173



Multiple causes are responsible for **MISSING DATA OR MISSING OBSERVATIONS**:

- The attribute's value is not always measurable/observable
- The attribute was non considered to be relevant when first collecting the data
- Misunderstanding concerning the guidelines to save the data
- Bad functioning of measuring and/or of saving devices
- Inconsistent values with respect to other attributes

The **MISSING DATA OR MISSING OBSERVATIONS PROBLEM** must be addressed, if any, before starting any Data Mining study.

WHICH are the **OPTIONS TO SOLVE** the **MISSING DATA OR MISSING OBSERVATIONS PROBLEM**?



OPTIONS TO SOLVE THE MISSING DATA OR MISSING OBSERVATIONS PROBLEM

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✓ RECORD REMOVAL

Assume 1% data is missing for each field

A record with **9 ATTRIBUTES** has a **PROBABILITY OF NO MISSING DATA** given by $(0.99)^9 = \mathbf{0.9135}$.

If the **NUMBER OF ATTRIBUTES** increases to **100** that **PROBABILITY RISES TO** $(0.99)^{100} = \mathbf{0.366}$.

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✓ **MANUAL IMPUTATION**

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✓ MANUAL IMPUTATION

Extremely time consuming.

OPTIONS TO SOLVE THE MISSING DATA OR MISSING OBSERVATIONS PROBLEM

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✓ **GLOBAL CONSTANT**

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✓ MODE REPLACEMENT

KS = 3/9
OH = 1/9
OK = 1/9
NJ = 1/9
... =

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KS	=	3/9	MODE
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✓ MEAN REPLACEMENT

$$(3 + 3 + 5 + 7 + 3 + 7 + 6 + 4 + 5) / 9 = 4.77$$

MEAN

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✓ **CONDITIONAL MEAN REPLACEMENT**

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✓ CONDITIONAL MEAN REPLACEMENT

$$(3 + 4) / 2 = 3.5$$

CONDITIONAL MEAN

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✓ **MOST PROBABLE**

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✓ **MOST PROBABLE**

OPTIONS TO SOLVE THE MISSING DATA OR MISSING OBSERVATIONS PROBLEM



Day Mins	Intl Calls
265.1	3
161.6	3
243.4	5
299.4	7
166.7	3

218.2	7
157	6
184.5	4
258.6	5

✓ **MOST PROBABLE**

$$\text{IntlCalls} = \alpha_0 + \alpha_1 \cdot \text{DayMins} + \varepsilon$$

OPTIONS TO SOLVE THE MISSING DATA OR MISSING OBSERVATIONS PROBLEM

Day Mins	Intl Calls
265.1	3
161.6	3
243.4	5
299.4	7
166.7	3
218.2	7
157	6
184.5	4
258.6	5

✓ **MOST PROBABLE**

$$\text{IntlCalls} = 2.23 + 0.012 \cdot \text{Day Mins}$$

OPTIONS TO SOLVE THE MISSING DATA OR MISSING OBSERVATIONS PROBLEM

Day Mins	Intl Calls
265.1	3
161.6	3
243.4	5
299.4	7
166.7	3
223.4	
218.2	7
157	6
184.5	4
258.6	5

✓ **MOST PROBABLE**

$$\text{Intl Calls} = 2.23 + 0.012 \cdot \text{Day Mins}$$

$$\text{Intl Calls} = 2.23 + 0.012 \cdot 223.4$$

OPTIONS TO SOLVE THE MISSING DATA OR MISSING OBSERVATIONS PROBLEM

Day Mins	Intl Calls
265.1	3
161.6	3
243.4	5
299.4	7
166.7	3
223.4	
218.2	7
157	6
184.5	4
258.6	5

✓ **MOST PROBABLE**

$$\text{Intl Calls} = 2.23 + 0.012 \cdot \text{Day Mins}$$

$$\text{Intl Calls} = 2.23 + 0.012 \cdot 223.4$$

OPTIONS TO SOLVE THE MISSING DATA OR MISSING OBSERVATIONS PROBLEM

Day Mins	Intl Calls
265.1	3
161.6	3
243.4	5
299.4	7
166.7	3
223.4	4.91
218.2	7
157	6
184.5	4
258.6	5

✓ **MOST PROBABLE**

$$\text{IntlCalls} = 2.23 + 0.012 \cdot \text{Day Mins}$$

$$4.91 = 2.23 + 0.012 \cdot 223.4$$

OPTIONS TO SOLVE THE MISSING DATA OR MISSING OBSERVATIONS PROBLEM

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265.1	3
161.6	3
243.4	5
299.4	7
166.7	3
223.4	4.91
218.2	7
157	6
184.5	4
258.6	5

✓ MOST PROBABLE

- More than one Attribute can be used with the linear regression model.
- More complex models can be used.
- It is possible to manage the case of Qualitative Attributes also, appropriate models.